Max Booth

CONTACT

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The Johns Hopkins University
Baltimore, MD 21211

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RESEARCH

Mathematician interested in Discrete Stochastic Processes, Spectral Theory, and Applied Mathematics in Data Science & Machine Learning. Previously applied techniques in Mathematics & Biostatistics to construct and validate predictive mathematical models of cancer progression and neurological development at the Johns Hopkins University School of Medicine.

EDUCATION

Johns Hopkins University, Ph.D. in Applied Mathematics and Statistics — Current

University of Utah, B.S. in Applied Mathematics — 2023
Summa Cum Laude

EXPERIENCE

Teaching Assistant — 2024—
Johns Hopkins University

- *Honors Discrete Mathematics*— FA 2024

Research Assistant — 2022—2024
The Johns Hopkins University School of Medicine
Advisors: Elana Fertig, Genevieve Stein-O'Brien

Undergraduate Teaching Assistant — 2019—2022
University of Utah
Supervisor: Lisa Penfold

- *Tutor, Foundations of Analysis Tutor*
- *Grader: MATH 1050, 1080, 1210, 1220, 1310, 1320, 2200, 2210, 2250, 2270, 3150, 3210, & 3220*

Private Tutor— Mathematics — 2017—2023
Salt Lake City, UT

- *College Algebra, Calculus, Linear Algebra, ODEs, PDEs, Discrete Mathematics, Foundations of Analysis I, & Foundations of Analysis II*

PUBLICATIONS & PREPRINTS

[1] **Digitize your Biology! Modeling multicellular systems through interpretable cell behavior**
Submitted (J. Johnson, G. Stein O'Brien, M. Booth, R. Heiland, F. Kurtoglu et al.)
Available at bioRXiV: https://doi.org/10.1101/2023.09.17.557982

PROJECTS & RESEACH

PhysiCell Workshop and Hackathon — 2023
Indiana University Bloomington
Random Walks ... with Modulated Responses to Physical Barriers

Research Internship 2022
The Johns Hopkins University School of Medicine
Supervisor: Elana Fertig

AWARDS

President's Scholarship 2019—2023
University of Utah

CONFERENCES ATTENDED

IMAG/MSM Past2Future Meeting 2023